

EE 512: CMOS Analog IC Design

Course Code	EE 512
Course Name	CMOS Analog IC Design
L-T-P-C	3-0-2-4
Prerequisites	Network theory (EE203), Signals and System (IC260), Control Theory (EE301) and the Instructors consent.
Intended for	3rd and 4th Year UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **MOS device models and short channel effects:** [4 Lectures]
 - MOSFET level 1 and level 2 models, threshold voltage model, capacitance model, mobility model, MOSFET basics, small-signal model derivation for a single transistor amplifier
 - Process, voltage, temperature (PVT) dependency and analog layout design essential considerations
- **Single stage amplifiers:** [4 Lectures]
 - Basic concept, Common source stage: with resistive load, with diode connected load, with current- source load, with triode load, with source degeneration
 - Source follower (common-drain) and common gate with various loads
- **CMOS Differential amplifiers:** [4 Lectures]
 - Single ended differential operation, basic differential pair (qualitative and quantitative analysis), common mode response, differential pair with MOS loads and Gilbert cell multiplier
 - Concept of matching transistors for analog layout, analog layout techniques for differential amplifier
- **CMOS Current mirrors:** [4 Lectures]
 - Scheme and implementation: basic current mirrors, cascode current mirrors and active current mirrors with large and small signal analysis
 - Understanding of common-mode properties
 - Analog layout making techniques for current mirrors
- **CMOS amplifier Frequency response:** [4 Lectures]
 - Miller effect, common source (CS), common gate (CG), common drain (CD) stages and cascode stage
 - Analog layout techniques for MIM, MOM and fringe capacitor
- **Noise analysis of the CMOS amplifier circuits:** [4 Lectures]
 - Types of noise, significance of flicker and thermal.
 - Analysis and representation of noise in single stage amplifiers: CG, CS, CD (source follower) and cascode stage and noise in differential pairs.
- **Feedback:** [4 Lectures]
 - Feedback topologies (voltage-voltage, current-voltage, voltage-current, current-voltage) and the noise and the loading effect analysis.
- **Design of the CMOS operational amplifiers:** [9 Lectures]
 - One-stage opamps and two stage opamps,
 - Gain boosting techniques, folded cascode, telescopic amplifier and common mode feedback (CMFB) amplifier
 - Three stage opamp architectures, opamp specifications analysis, Design of high speed and high gain amplifiers
- **Stability and frequency compensation:** [4 Lectures]
 - Specification analysis, multi-pole system, three stage opamp, phase margin
 - Frequency compensation, pole-zero doublet analysis
- **Analog layout techniques:** [1 Lectures]
 - Design rule check (DRC), layout versus schematic (LVS) and antenna effects
 - Design of pad-ring and gds file generation

Text Books

1. Behzad Razavi, **Design of Analog CMOS Integrated Circuits**, McGraw Hill Education, 2000.

References

1. Phillip Allen and Douglas R. Holberg, **CMOS Analog Circuit Design**, 3rd Edition, 2011.
2. Yannis Tsividis, **Operation and Modeling of the MOS Transistor**, 2nd Edition, Oxford University Press, 2003.
3. A.S. Sedra and K.C. Smith, Adapted by A.N. Chandorkar, **Microelectronic Circuits-Theory & Applications**, 6th Edition, Oxford, 2013.